UNIVERSITY MOHAMED BOUDIAF OF M'SILA
Faculty of Mathematics and Computer Science
Department of Computer Science

TP N°1 — Laboratory Report

Introduction to R

FDTA — Fouille de Données, Techniques et Applications

Prepared by: Younes Barkat
Level: Master 01 — Artificial Intelligence
Supervised by: Dr. Taher MEHANNI
Academic Year: 2025 / 2026

TP N°1: Introduction to R

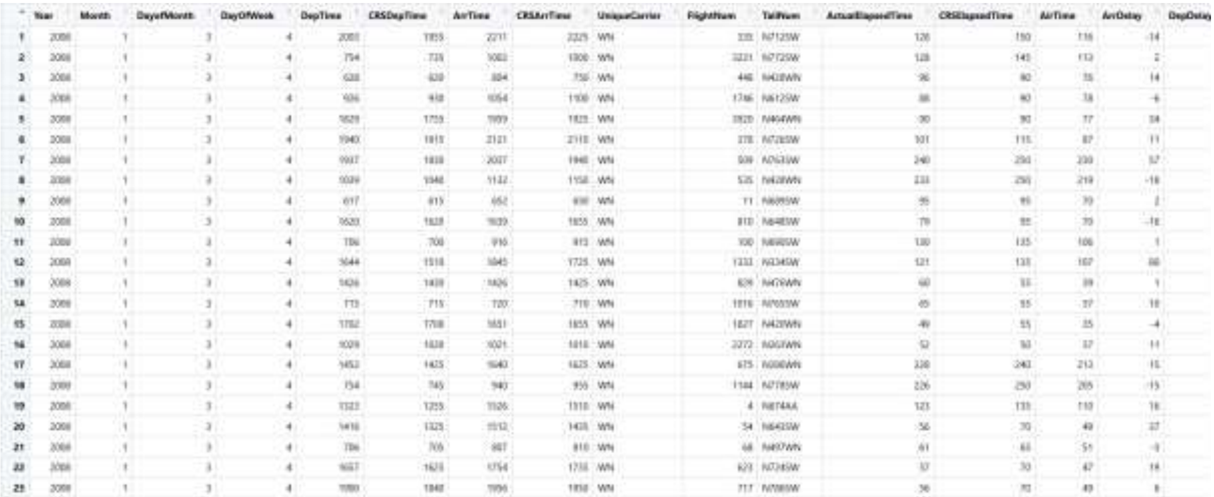
Objectives

This lab introduces core R programming skills applied to the 2008.csv U.S. domestic flights dataset (29 variables, ~7 million records). Tasks cover: data import, column access, Boolean filtering, subset creation, aggregate statistics (sum, mean), frequency tables, and graphical visualization.

Step 2: Importing the Dataset

```
myDf <- read.csv("C:/Users/Younes/Desktop/TP_FDDTIA/TP1/2008.csv")
```

```
> myDf<-read.csv("C:/Users/Younes/Desktop/TP_FDDTA/TP1/2008.csv")
```



read.csv() parses the CSV file into a data frame, making all 29 columns immediately accessible. The RStudio viewer confirms correct import — numeric fields (DepTime, ArrDelay) and categorical fields (UniqueCarrier, Origin) are properly typed.

Step 3: Viewing the First 6 Rows

```
head(myDf)
```

```
> head(myDf)
  Year Month DayofMonth DayOfWeek DepTime CRSDepTime ArrTime CRSArrTime UniqueCarrier FlightNum
1 2008     1          3         4    2003       1955    2211       2225            WN       335
2 2008     1          3         4     754        735    1002       1000            WN      3231
3 2008     1          3         4     628        620     804        750            WN       448
4 2008     1          3         4     926        930    1054       1100            WN      1746
5 2008     1          3         4    1829       1755    1959       1925            WN      3920
6 2008     1          3         4    1940       1915    2121       2110            WN       378
  TailNum ActualElapsedTime CRSElapsedTime AirTime ArrDelay DepDelay Origin Dest Distance TaxiIn
1  N712SW               128            150     116      -14        8    IAD  TPA      810      4
2  N772SW               128            145     113        2       19    IAD  TPA      810      5
3  N428WN                96             90      76       14        8    IND  BWI      515      3
4  N612SW                88             90      78       -6       -4    IND  BWI      515      3
5  N464WN                90             90      77       34       34    IND  BWI      515      3
6  N726SW               101            115      87       11       25    IND  JAX      688      4
  TaxiOut Cancelled CancellationCode Diverted CarrierDelay WeatherDelay NASDelay SecurityDelay
1       8         0                         0           NA           NA       NA            NA
2      10         0                         0           NA           NA       NA            NA
3      17         0                         0           NA           NA       NA            NA
4       7         0                         0           NA           NA       NA            NA
5      10         0                         0            2            0        0             0
6      10         0                         0           NA           NA       NA            NA
  LateAircraftDelay
1                NA
2                NA
3                NA
4                NA
5                32
6                NA
```

head() retrieves the first six records for a fast structural preview, confirming column alignment across all 29 variables — including time fields, carrier codes, delay metrics, and cancellation flags — without printing millions of rows.

Step 4: First 6 Values of the Origin Attribute

```
head(myDf$Origin)
```

```
> head(myDf$Origin)
[1] "IAD" "IAD" "IND" "IND" "IND" "IND"
```

The $ operator isolates the Origin column. Result ["IAD" "IAD" "IND" "IND" "IND" "IND"] shows the first flights departed from Washington Dulles (IAD) and Indianapolis (IND).

Step 5: Last 6 Values of the Origin Attribute

```
tail(myDf$Origin)
```

```
> tail(myDf$Origin)
[1] "SAV" "ATL" "ATL" "PBI" "IAD" "SAT"
```

tail() retrieves the final six entries: ["SAV" "ATL" "ATL" "PBI" "IAD" "SAT"], confirming geographic diversity across the dataset's end records.

Step 6: Boolean Check: Origin Equals "IND"

```
head(myDf$Origin == "IND")
```

```
> head(myDf$Origin=="IND")
[1] FALSE FALSE  TRUE  TRUE  TRUE  TRUE
```

A logical vector is produced (TRUE where Origin = "IND"). Result FALSE FALSE TRUE TRUE TRUE TRUE confirms rows 3–6 are IND flights. This Boolean indexing underpins all subsequent filtering operations.

Step 7: Total Departures from "IND"

```
sum(myDf$Origin == "IND")
```

```
> sum(myDf$Origin=="IND")
[1] 42750
```

sum() coerces TRUE→1, FALSE→0, counting 42,750 flights departing Indianapolis International Airport during 2008.

Step 8: Total Arrivals at "IND"

```
sum(myDf$Dest == "IND")
```

```
> sum(myDf$Dest=="IND")
[1] 42732
```

42,732 inbound flights to IND — nearly matching departures, consistent with balanced hub scheduling where arrivals and departures remain symmetrical over a full year.

Step 9: Total Departures from "ORD"

```
sum(myDf$Origin == "ORD")
```

```
> sum(myDf$Origin=="ORD")
[1] 350380
```

350,380 departures from Chicago O'Hare — roughly 8× IND's volume, reflecting O'Hare's dominance as one of the busiest aviation hubs in the United States.

Step 10: Total Arrivals at "ORD"

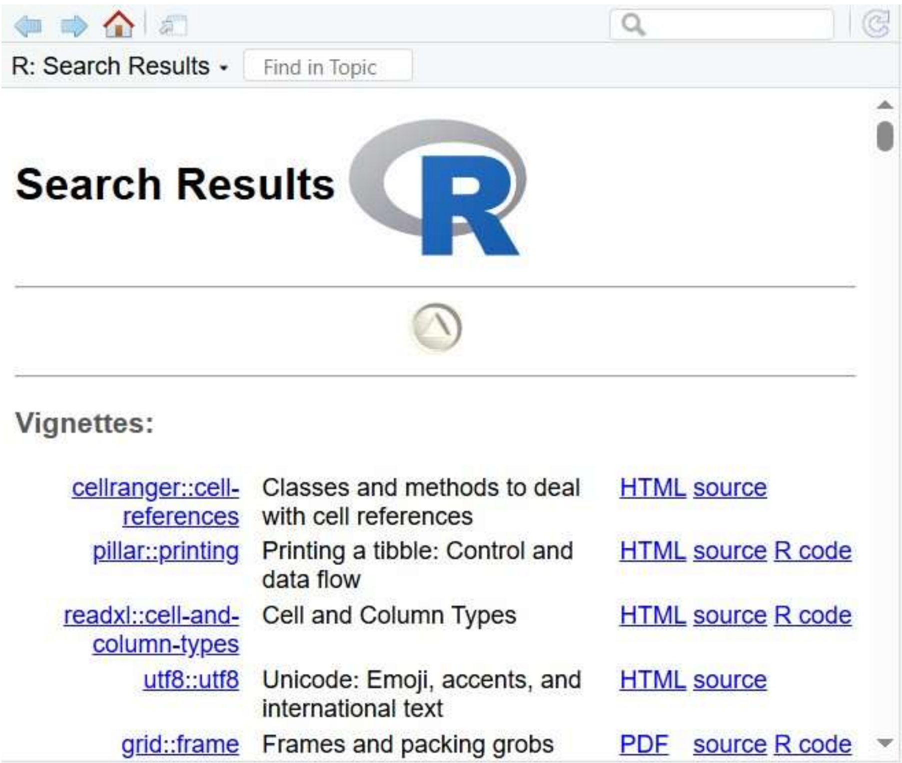
```
sum(myDf$Dest == "ORD")
```

The symmetric arrival count at O'Hare complements Step 9. Near-equal arrival/departure totals confirm the balanced carrier scheduling strategy at this major hub.

Step 11: R Documentation Search: ??AND

```
??AND
```

```
> ??AND
```



??AND triggers a broad package-wide documentation search. Since AND is not a native R function (R uses & for element-wise conjunction), the search returns general vignettes, illustrating the distinction between operators and named functions in R.

Step 12: Flights from "IND" to "ORD"

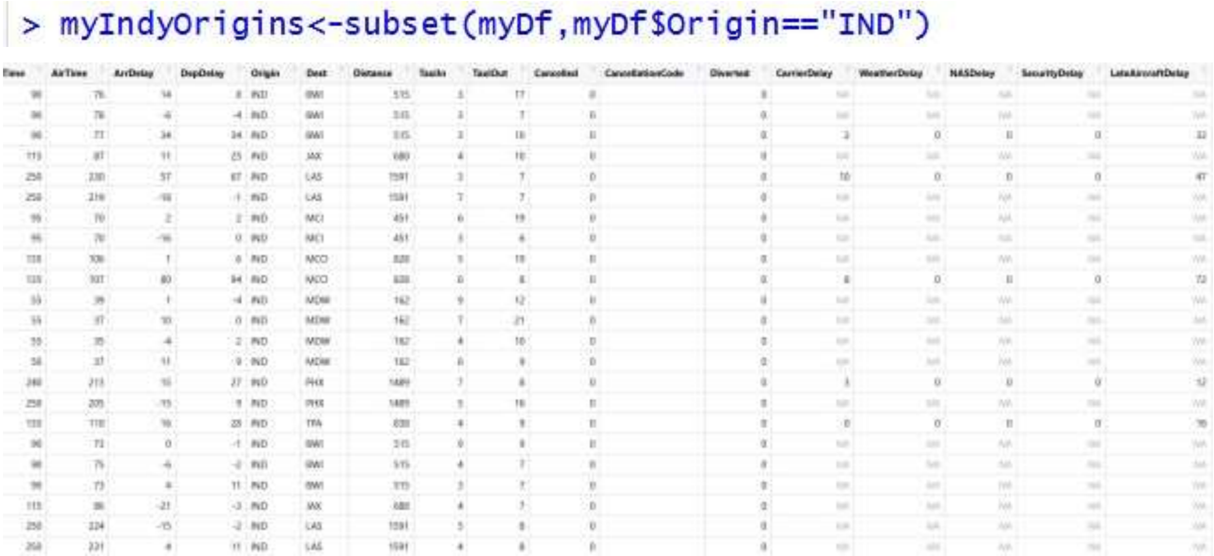
```
sum(myDf$Origin == "IND" & myDf$Dest == "ORD")
```

```
> sum(myDf$Origin=="IND" & myDf$Dest=="ORD")
[1] 4102
```

The & operator requires both conditions to be satisfied simultaneously. 4,102 IND→ORD flights operated in 2008, making it one of Indianapolis's primary served corridors.

Step 13: Subset: IND-Origin Flights

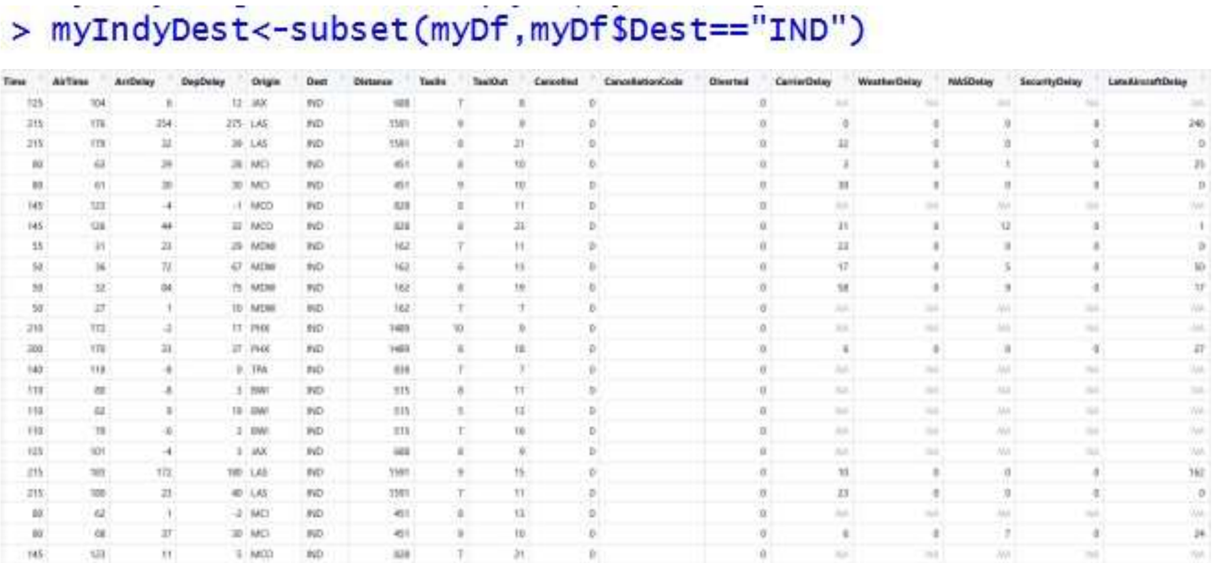
```
myIndyOrigins <- subset(myDf, myDf$Origin == "IND")
```



subset() creates a dedicated data frame with only IND-departing rows (42,750 rows × 29 cols), enabling focused analysis without re-filtering the 7M-row master dataset on every operation.

Step 14: Subset: IND-Destination Flights

```
myIndyDest <- subset(myDf, myDf$Dest == "IND")
```



A parallel subset captures 42,732 IND-arriving flights into myIndyDest, enabling direct inbound vs. outbound comparison of scheduling patterns and delay distributions.

Step 15: Preview of myIndyOrigins

```r
head(myIndyOrigins)
```

```
> head(myIndyOrigins)
  Year Month DayofMonth DayOfWeek DepTime CRSDepTime ArrTime CRSArrTime UniqueCarrier
3 2008     1          3         4     628        620     804        750            WN
4 2008     1          3         4     926        930    1054       1100            WN
5 2008     1          3         4    1829       1755    1959       1925            WN
6 2008     1          3         4    1940       1915    2121       2110            WN
7 2008     1          3         4    1937       1830    2037       1940            WN
8 2008     1          3         4    1039       1040    1132       1150            WN
  FlightNum TailNum ActualElapsedTime CRSElapsedTime AirTime ArrDelay DepDelay Origin Dest
3       448  N428WN                96             90      76       14        8    IND  BWI
4      1746  N612SW                88             90      78       -6       -4    IND  BWI
5      3920  N464WN                90             90      77       34       34    IND  BWI
6       378  N726SW               101            115      87       11       25    IND  JAX
7       509  N763SW               240            250     230       57       67    IND  LAS
8       535  N428WN               233            250     219      -18       -1    IND  LAS
  Distance TaxiIn TaxiOut Cancelled CancellationCode Diverted CarrierDelay WeatherDelay
3      515      3      17         0                         0           NA           NA
4      515      3       7         0                         0           NA           NA
5      515      3      10         0                         0            2            0
6      688      4      10         0                         0           NA           NA
7     1591      3       7         0                         0           10            0
8     1591      7       7         0                         0           NA           NA
  NASDelay SecurityDelay LateAircraftDelay
3       NA            NA                NA
4       NA            NA                NA
5        0             0                32
6       NA            NA                NA
7        0             0                47
8       NA            NA                NA
```

All six rows confirm Origin = "IND". Destinations include BWI and JAX with departure times and delay values visible, validating the subset filter correctness.

Step 16: Preview of myIndyDest

```r
head(myIndyDest)
```

```
> head(myIndyDest)
    Year Month DayofMonth DayOfWeek DepTime CRSDepTime ArrTime CRSArrTime UniqueCarrier FlightNum TailNum
72  2008     1          3         4     802        750    1001        955            WN      2272  N263WN
136 2008     1          3         4    2255       1820     509         55            WN      1924  N761RR
137 2008     1          3         4    1129       1050    1757       1725            WN      3920  N464WN
501 2008     1          3         4    1038       1010    1259       1230            WN         4  N674AA
502 2008     1          3         4    1920       1850    2140       2110            WN       321  N742SW
605 2008     1          3         4    1139       1140    1401       1405            WN       829  N476WN
    ActualElapsedTime CRSElapsedTime AirTime ArrDelay DepDelay Origin Dest Distance TaxiIn TaxiOut
72                119            125     104        6       12    JAX  IND      688      7       8
136               194            215     176      254      275    LAS  IND     1591      9       9
137               208            215     179       32       39    LAS  IND     1591      8      21
501                81             80      63       29       28    MCI  IND      451      8      10
502                80             80      61       30       30    MCI  IND      451      9      10
605               142            145     123       -4       -1    MCO  IND      828      8      11
    Cancelled CancellationCode Diverted CarrierDelay WeatherDelay NASDelay SecurityDelay
72          0                         0           NA           NA       NA            NA
136         0                         0            0            0        0             8
137         0                         0           32            0        0             0
501         0                         0            3            0        1             0
502         0                         0           30            0        0             0
605         0                         0           NA           NA       NA            NA
    LateAircraftDelay
72                 NA
136               246
137                 0
501                25
502                 0
605                NA
```

All rows confirm Dest = "IND". Origins span LAS, MCI, and JAX; one flight shows a 275-minute departure delay, highlighting the range of punctuality among airports feeding into Indianapolis.

Step 17: Monthly Flight Frequency Table

```
table(myIndyOrigins$Month)
```

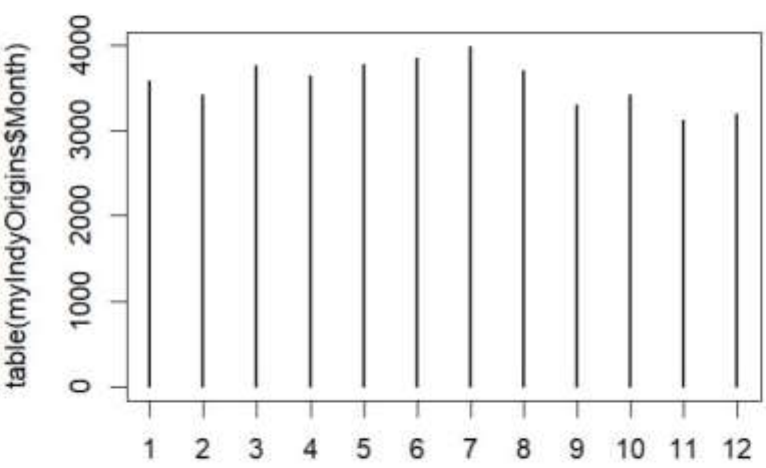
```
> table(myIndyOrigins$Month)

   1    2    3    4    5    6    7    8    9   10   11   12 
3580 3414 3764 3644 3768 3852 3986 3700 3300 3418 3126 3198 
```

table() counts flights per month. July peaks at 3,986 and November dips to 3,126, reflecting stronger summer demand and reduced autumn leisure travel from IND.

Step 18: Monthly Distribution Plot

```
plot(table(myIndyOrigins$Month))
```

```
> plot(table(myIndyOrigins$Month))
```



Passing a table object to plot() generates a bar chart. The graph visually confirms the summer peak (months 6–7) and autumn dip (months 9–11), consistent with U.S. domestic air travel seasonality.

Step 19: Total Flights from "TUP"

```
sum(myDf$Origin == "TUP")
```

```
> sum(myDf$Origin=="TUP")
[1] 10
```

Only 10 flights departed from Tupelo Regional (TUP) in 2008, illustrating the extreme volume skewness where a few major hubs dominate national traffic while hundreds of small airports handle negligible volumes.

Step 20: Proportion of TUP-Origin Flights

```
mean(myDf$Origin == "TUP")
```

```
> mean(myDf$Origin=="TUP")
[1] 1.426589e-06
```

mean() on a logical vector computes relative frequency. Result 1.43e-06 (~0.000143%) confirms TUP's infinitesimally small share of national domestic traffic in 2008.

Step 21: Subset TUP-Origin Flights & Average Departure Delay

```
myTUPOrigins <- subset(myDf, myDf$Origin == "TUP")
```

```
> myTUPOrigins<-subset(myDf, myDf$Origin=="TUP")
```

```
mean(myTUPOrigins$DepDelay)
```

```
> mean(myTUPOrigins$DepDelay)
[1] -3.8
```

The 10 TUP flights are isolated into myTUPOrigins; carrier EV (ExpressJet) is typical of regional feeders. Mean departure delay = −3.8 minutes — flights leave slightly early on average due to the complete absence of congestion at this micro-airport.

Step 23: Departure Time Analysis for IND Flights (Steps 23–26)

```
sum(myIndyOrigins$DepTime < 600,  na.rm=TRUE)  # Before 06:00 → Result: 692
```

```
> sum(myIndyOrigins$DepTime<600, na.rm=TRUE)
[1] 692
```

```
sum(myIndyOrigins$DepTime < 1200, na.rm=TRUE)  # Before 12:00 → Result: 18,705
```

```
> sum(myIndyOrigins$DepTime<1200, na.rm=TRUE)
[1] 18705
```

```
sum(myIndyOrigins$DepTime < 1800, na.rm=TRUE)  # Before 18:00 → Result: 35,004
```

```
> sum(myIndyOrigins$DepTime<1800, na.rm=TRUE)
[1] 35004
```

```
sum(myIndyOrigins$DepTime < 2400, na.rm=TRUE)  # Before 24:00 → Result: 42,011
```

```
> sum(myIndyOrigins$DepTime<2400, na.rm=TRUE)
[1] 42011
```

DepTime is encoded as HHMM integers; na.rm=TRUE excludes cancelled/missing records. Results show 692 early-morning departures (<06:00), 18,705 before noon (~44%), 35,004 before 18:00 (~82%), and 42,011 within a valid time range. The heavy front-loading before noon reflects standard airline scheduling to maximize aircraft daily utilization.

Step 27: Flights with Missing Departure Time

```
sum(is.na(myIndyOrigins$DepTime))
```

```
> sum(myDf$Dest=="LAX")
[1] 215685
```

is.na() identifies genuinely undefined (NA) records — these are flights that were cancelled or had no departure time logged. They must be excluded from temporal analysis to avoid biased aggregate computations.

Step 28: Total Arrivals at "LAX"

```
sum(myDf$Dest == "LAX")
```

```
> sum(myDf$Origin=="ATL" & myDf$Dest=="LAX")
[1] 5406
```

215,685 flights landed at Los Angeles International (LAX) in 2008, placing it among the top destination airports and reflecting its role as Southern California's primary gateway.

Step 29: Flights from "ATL" to "LAX"

```
sum(myDf$Origin == "ATL" & myDf$Dest == "LAX")
```

```
> myATLandLAX<-subset(myDf, myDf$Origin=="ATL" & myDf$Dest=="LAX")
```

5,406 flights operated the ATL→LAX corridor — roughly 15 per day — one of the longest and most competitive domestic routes connecting the East and West coasts.

Step 30: Subset: ATL-to-LAX Flights

```
myATLandLAX <- subset(myDf, myDf$Origin == "ATL" & myDf$Dest == "LAX")
```

Time	AirTime	ArrDelay	DepDelay	Origin	Dest	Distance	TaxiIn	TaxiOut	Cancelled	CancellationCode	Diverted	CarrierDelay	WeatherDelay	NASDelay	SecurityDelay	LateAircraftDelay
285	267	8	7	ATL	LAX	1946	8	15	0		0	NA	NA	NA	NA	NA
300	265	-9	-3	ATL	LAX	1946	14	20	0		0	NA	NA	NA	NA	NA
291	268	1	-4	ATL	LAX	1946	7	21	0		0	NA	NA	NA	NA	NA
304	NA	NA	NA	ATL	LAX	1946	NA	NA	1	A	0	NA	NA	NA	NA	NA
297	268	237	7	ATL	LAX	1946	7	351	0		0	0	7	230	0	0
298	NA	NA	NA	ATL	LAX	1946	NA	NA	1	B	0	NA	NA	NA	NA	NA
296	268	-11	8	ATL	LAX	1946	5	14	0		0	NA	NA	NA	NA	NA
305	NA	NA	NA	ATL	LAX	1946	NA	NA	1	A	0	NA	NA	NA	NA	NA
300	265	-6	0	ATL	LAX	1946	8	22	0		0	NA	NA	NA	NA	NA
299	261	34	25	ATL	LAX	1946	10	29	0		0	25	0	9	0	0
295	257	4	6	ATL	LAX	1946	7	25	0		0	NA	NA	NA	NA	NA
285	261	-8	-2	ATL	LAX	1946	6	12	0		0	NA	NA	NA	NA	NA
291	271	16	12	ATL	LAX	1946	8	16	0		0	12	0	4	0	0
304	265	3	7	ATL	LAX	1946	15	20	0		0	NA	NA	NA	NA	NA
297	266	22	7	ATL	LAX	1946	18	32	0		0	7	0	15	0	0
298	264	-11	-1	ATL	LAX	1946	6	18	0		0	NA	NA	NA	NA	NA
299	265	3	1	ATL	LAX	1946	8	25	0		0	NA	NA	NA	NA	NA
300	271	-22	-2	ATL	LAX	1946	6	23	0		0	NA	NA	NA	NA	NA
315	268	-7	-1	ATL	LAX	1946	20	23	0		0	NA	NA	NA	NA	NA
291	263	54	53	ATL	LAX	1946	11	18	0		0	53	0	1	0	0
289	264	26	14	ATL	LAX	1946	14	29	0		0	12	0	12	0	2
285	268	32	28	ATL	LAX	1946	4	19	0		0	0	7	0	0	25
305	264	24	25	ATL	LAX	1946	15	25	0		0	0	24	0	0	0

The combined filter extracts all 5,406 ATL→LAX records into a dedicated data frame for route-specific analysis of delays and scheduling without querying the full 7M-row dataset repeatedly.

Step 31: ATL→LAX Flights Departing Before Noon

```
sum(myATLandLAX$DepTime < 1200, na.rm=TRUE)
```

```
> sum(myATLandLAX$DepTime<1200, na.rm=TRUE)
[1] 2133
```

2,133 of 5,406 ATL→LAX flights (~39%) depart before noon. Given the ~4.5-hour flight time, morning departures allow passengers to arrive in LA by mid-afternoon, maximizing usable West Coast business hours.

Step 32: Departure Time Frequency Table — ATL→LAX

```
table(myATLandLAX$DepTime)
```

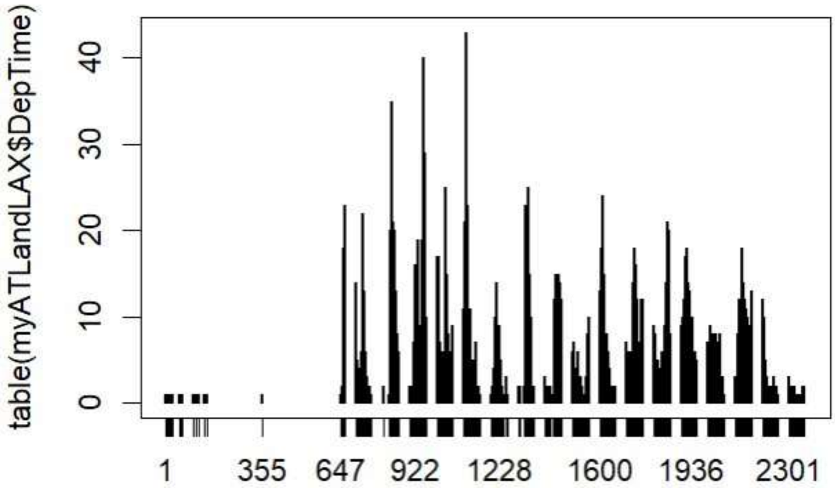
```
> table(myATLandLAX$DepTime)

   1    2   12   16   19   25   27   48   49   51   57   58  101  109  123  144  152  355  647  652  653
   1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2
 654  655  656  657  658  659  700  701  702  704  705  707  708  709  711  712  713  714  715  716  718
   2    5   11   16   18   23   14    7    6    4    5    2    3    1    2    1    4    2    2    1    2
 719  720  721  722  723  724  725  726  727  728  729  730  731  732  733  734  735  736  737  738  739
   2    4    4    2    6    4   11    5    9   14   22   13    8   11    8    5    6    4    4    6    1
 740  741  742  743  746  747  749  753  755  801  823  826  827  828  829  830  831  832  833  834  835
   3    3    3    1    2    1    1    2    1    2    1    1    2    3    7   12   20   13   34   35   24
 836  837  838  839  840  841  842  843  844  845  846  847  848  849  850  851  852  853  854  855  856
  25    8    9   21   19   10    7   10   20   14   11    6    6   13    2    5    8    3    2    3    4
 857  858  859  900  901  904  906  908  911  912  914  915  916  917  918  919  920  921  922  923  924
   1    6    3    2    2    2    1    1    1    1    1    1    7    7    6   16   15    8   11   13   11
 925  926  927  928  929  930  931  932  933  934  935  936  937  938  939  940  941  942  943  944  945
  11    6   11    7   10   19    5    8    9    7    8    5    6    6    3    4    7    5    2    3    9
 946  947  948  949  950  951  952  953  954  955  956  957  958  959 1000 1001 1002 1003 1004 1005 1006
  11   12   19   25   40   26   24   24   29   27   28   17   23   10   17    9   11    7   11   17    7
1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027
   8    9    5    4    6    7    1    1    4    6    2    3    3    3    1    4    1    2    1    2    5
1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044 1045 1046 1047 1048
   6    5   12    7   22   20   25   13   12   15   15    9    5    5    4    8    6    6    3    4    4
1049 1050 1051 1052 1053 1054 1055 1056 1057 1058 1059 1100 1101 1102 1103 1104 1105 1106 1107 1108 1109
   5    1    2    2    1    1    4    9    6    5    4   11    8    4    7   21   21   32   34   36   43
1110 1111 1112 1113 1114 1115 1116 1117 1118 1119 1120 1121 1122 1123 1124 1125 1126 1127 1128 1129 1130
  26   15   23   15   11   12   11    8   10    6   11    8   11    2   10    4    3    5    1    5    1
1131 1132 1133 1134 1135 1136 1137 1138 1139 1140 1141 1142 1143 1145 1146 1147 1148 1154 1155 1156 1159
   4    4    5    3    3    4    2    2    1    3    2    3    7    2    1    1    1    2    1    2    1
1201 1205 1206 1208 1210 1211 1212 1213 1214 1215 1216 1217 1218 1219 1220 1221 1222 1223 1224 1225 1226
   1    1    2    2    1    1    4    7    6   10    3    3    4    2    5   14    7    5    6    7    6
1227 1228 1229 1230 1231 1232 1233 1234 1235 1236 1237 1239 1240 1241 1242 1243 1246 1248 1254 1255 1256
   9    7    4    9    5    7    4    3    2    5    2    2    2    1    1    1    1    1    2    3    3
1257 1259 1302 1308 1309 1320 1321 1323 1324 1325 1326 1327 1328 1329 1330 1331 1332 1333 1334 1335 1336
   1    1    2    1    2    1    1    1    2    6   20   18   23   19   23   17   17   11    5   11   12
1337 1338 1339 1340 1341 1342 1343 1344 1345 1346 1347 1348 1349 1350 1351 1352 1353 1354 1355 1356 1357
  11   14   25   15   10    6   10    7    5   10    2    4    2    1    1    2    2    2    2    1    2
1359 1400 1401 1407 1408 1409 1410 1414 1418 1419 1421 1428 1429 1430 1431 1432 1433 1434 1435 1436 1437
   1    3    1    2    1    2    1    1    2    1    2    1    1    1    1    1    2    4    6   12   10
```

table() enumerates every distinct departure time. The dense output shows hundreds of unique slots, with certain morning windows exceeding a count of 20, reflecting deliberate schedule clustering at peak demand hours.

Step 33: Departure Time Distribution Plot — ATL→LAX

```
plot(table(myATLandLAX$DepTime))
```

```
> plot(table(myATLandLAX$DepTime))
```



The bar chart reveals a bimodal distribution with peaks around 08:00–10:00 and 12:00–13:00. Minimal activity before 06:00 and after 22:00 reflects airport noise restrictions and crew scheduling constraints common at major U.S. airports.

Conclusion

This laboratory session demonstrated R's complete data analysis pipeline applied to over seven million flight records: CSV ingestion, column-level access, Boolean filtering, subset construction, aggregate statistics, frequency tables, and graphical output.

Key findings: O'Hare (ORD) dominates with 350,380 departures vs Indianapolis's 42,750; 44% of IND flights depart before noon; TUP operates only 10 flights with a −3.8 min average delay; and the ATL→LAX corridor exhibits a clear bimodal scheduling pattern. These results confirm that even elementary R functions can extract actionable intelligence from large real-world datasets.